Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider a discrete memoryless source X that emits symbols from the alphabet $\{a, b, c\}$ according to the following p.m.f.:

symbol	Pr.
a	0.5
b	0.4
c	0.1

   (a) Determine the entropy of the source and lower/upper bounds on the average code length of a Shannon code that encodes $N = 1$ symbols at the time.

   (b) Build a Huffman code for this source ($N = 1$) and compute its average code length. Compare such length with the bounds above.

   (c) Consider a code that encodes $N = 2$ symbols at the time. Revise the lower/upper bounds above, build a Huffman code and compare its length with the revised bounds.

   (d) Consider the following sequences of symbols: *bccb*. Compute the number of bits per symbol using the Huffman codes above ($N = 1, 2$) and compare it with the bounds. Comment the result.

   Solution

   (a) The entropy of the source is equal to $H(X) = 1.361$ bps. Let L_1 denote the average code length of a Shannon coding. This quantity is bounded as follows

$$H(X) \leq L_1 < H(X) + 1 \tag{1}$$

$$1.361 \leq L_1 < 2.361 \tag{2}$$

   (b) A Huffman code for the source X is given by

symbol	Pr.	code	length
a	0.5	1	1
b	0.4	00	2
c	0.1	01	2

   The average code length is equal to $L_{H,1} = 1.5$ bps. Huffman coding is optimal. Hence the following inequality holds:

$$H(X) \leq L_{H,1} \leq L_1 < H(X) + 1 \tag{3}$$

(c) When coding groups of $N = 2$ symbols, let L_2 denote the average code length of a Shannon coding. This quantity is bounded as follows

$$H(X) \leq L_2 < H(X) + 1/2 \tag{4}$$

A Huffman code for the source X when $N = 2$ is given by

symbol	Pr.	code	length
aa	0.25	11	2
ab	0.20	00	2
ac	0.05	10010	5
ba	0.20	01	2
bb	0.16	101	3
bc	0.04	10001	5
ca	0.05	10011	5
cb	0.04	100001	6
cc	0.01	100000	6

The average code length is equal to 2.78 bits / supersymbol, i.e., $L_{H,2} = 1.39$ bps.
Hence the following inequality holds:

$$H(X) \leq L_{H,2} \leq L_2 < H(X) + 1/2 \tag{5}$$

(d) When $N = 1$, encoding *bccb* results into the sequence of bits 00010100, i.e., $R_1 = 2$ bps. When $N = 2$, we obtain: 10001100001, i.e., $R_2 = 2.75$ bps.

- Although both R_1 and R_2 exceed their respective average code lengths, this should not come as a surprise. It is perfectly fine that for a specific sequence of symbols the bounds are not fulfilled, since they hold on average, over several realizations.
- For the same reason, it should not be considered an issue the fact that $R_2 > R_1$, although $L_{H,2} \leq L_{H,1}$.

2. **Question (5 pts)**

Discuss the design of an open-loop vocoder.

Solution

See slide set `MMSP_2.1_Speech coding`.

3. **Question (5 pts)**

(a) Illustrate the block diagram of a transform coder.

(b) Describe how to determine the optimal bit allocation, once the transform is given.

Solution

See slide set `MMSP_1.4_Transform_Coding`.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider a DPCM coder.
   - [T] [F] $1 \leq G_{DPCM} < G_{VQ}$, where G_{DPCM} and G_{VQ} denote, respectively, the coding gain of DPCM and vector quantization.
     TRUE.
   - [T] [F] The quantization error on the prediction residual is the same as the quantization error on the original samples.
     TRUE. $\tilde{x} - x = \tilde{d} - d$
   - [T] [F] The predictor is computed based on previous original samples, i.e., $\hat{x}(n) = f(x(n-1), x(n-2), \ldots)$
     FALSE. $\hat{x}(n) = f(\tilde{x}(n-1), \tilde{x}(n-2), \ldots)$. In the case of linear prediction, $\hat{x}(n) = \sum_{k=1}^{p} a_k \tilde{x}(n-k)$
   - [T] [F] In an Adaptive DPCM coder (ADPCM), it is possible to vary both the quantizer and the predictor.
     TRUE.

   (b) Consider an image coder based on transform coding
   - [T] [F] If $N > M$, then $R_N(D) \leq R_M(D)$, where $R_N(D)$ denotes the number of bits needed to encode the image at a target distortion D, when using $N \times N$ blocks.
     TRUE. Increasing the block size increases the coding efficiency, since it is possible to exploit inter-pixel correlation at longer distances.
   - [T] [F] The different color components in natural images (e.g., RGB components) are statistically independent.
     FALSE. Pixels values at the same location, but belonging to different color components, are statistically dependent. This dependency is due to the overlapping between the spectral response of the acquisition sensors. The degree of dependency is reduced (but not eliminated) by applying a color transform (e.g., RGB to YCbCr).
   - [T] [F] When adopting threshold coding, each image block is allocated the same number of bits.
     FALSE. The same quantizer is applied to all image blocks. However, since the signal is not stationary, this results in different bit budgets.

   (c) Consider a motion estimation/motion compensation algorithm.
   - [T] [F] Motion estimation is more computationally expensive than motion compensation.
     TRUE.
   - [T] [F] When increasing the block size, the energy of the prediction residuals increases.
     TRUE. Larger blocks induce a more coarse motion representation, thus leading to larger residuals.
   - [T] [F] Motion estimation/compensation at sub-pixel accuracy requires to interpolate both the current and the reference frame.
     FALSE. Only the reference frame needs to be interpolated.

5. **Question (Matlab - 10pts)**
   Load the files $f1.mat$ and $f2.mat$ which contain two consecutive frames of a video sequence.
   (a) Encode $f2$ with a PCM encoder. Use the following values for the step size Q = [2 4 8 16 32 64]. Plot the rate-distortion function, where rate is estimated based on the entropy of the quantized symbols, and distortion is measured through the PSNR.

(b) Encode $f2$ with a *closed-loop* DPCM encoder. Use the same step sizes as before. Plot the rate-distortion function in this case. Comment the result.

Solution

```
clear
close all
clc


% Load the files f1.mat and f2.mat which contain consecutive frames of a
% video sequence.
load f1.mat
load f2.mat

f1 = f1(:);
f2 = f2(:);


%Encode f2 with a PCM encoder. Use the following values for the step size
%Q = [2 4 8 16 32 64]. Plot the rate-distortion function, where rate is
%estimated based on the entropy of the quantized symbols, and the
%distortion measure is the PSNR.
Q = [2 4 8 16 32 64];
H_pcm = zeros(1,length(Q));
PSNR_pcm = H_pcm;
iq = 1;
for q = Q
    %PCM encoding
    f2_pcm = q*floor(f2/q)+q/2;

    %Rate estimation
    p = hist(f2_pcm,min(f2_pcm):max(f2_pcm));
    p = p(p~=0);
    p = p./sum(p);
    H_pcm(iq) = -sum(p.*log2(p));

    %Compute distortion
    PSNR_pcm(iq) = 10*log10(255^2/var(f2 - f2_pcm));

    iq = iq+1;
end

%Encode f2 with a closed-loop DPCM encoder.
%Use the same step sizes as before. Plot the rate-distortion function in
%this case. Comment the result.
H_dpcm = zeros(1,length(Q));
PSNR_dpcm = H_pcm;
iq = 1;
for q = Q
    %PCM encoding of f1
    f1_pcm = q*floor(f1/q)+q/2;

    %DPCM encoding
    d = f2 - f1_pcm;
    dq = q*floor(d/q)+q/2;
    f2_dpcm = f1_pcm+dq;

    %Rate estimation
    p = hist(dq,min(dq):max(dq));
    p = p(p~=0);
    p = p./sum(p);
    H_dpcm(iq) = -sum(p.*log2(p));

    %Compute distortion
    PSNR_dpcm(iq) = 10*log10(255^2/var(f2 - f2_dpcm));

    iq = iq+1;
end

%Plot
plot(H_pcm,PSNR_pcm,'b');
hold on
plot(H_dpcm,PSNR_dpcm,'r')
legend('PCM','DPCM');
xlabel('Rate [bps]');
ylabel('PSNR [dB]');
grid on


%COMMENT: The R-D function of DPCM coding outperforms PCM. This is due
%primarily to the difference between the p.m.f of the PCM quantized symbols
%and the p.m.f of the DPCM quantized residuals. In the latter case, the
%p.m.f has a sharp peak around zero, which makes entropy coding particularly
%efficient.
```